# Classical Field Theory — Exercise Sheet 24

*A higher Lax flow and operator domains*

**Hints and answers**

Use $D_x = \partial_x$ and the ordered commutator $[B, L] = BL - LB$. All operator calculations in the first problem are on $C_c^\infty(\mathbb{R}, \mathbb{C})$, with smooth real coefficient $u$. A displayed differential expression does not specify its Hilbert-space domain.

**Exercise 24.1 (Determining a fifth-order Lax flow).** Let $L = -D_x^2 + u$. For real constants $r_1, \dots, r_6$ consider

$$B_5 = 16D_x^5 + r_1 u D_x^3 + r_2 u' D_x^2 + (r_3 u'' + r_4 u^2) D_x + r_5 u''' + r_6 u u'.$$

Products with $u$ and its derivatives here mean multiplication before the indicated derivative of the test.

(a) Determine the constants so that $[B_5, L]$ is multiplication for every smooth $u$. Find the resulting scalar coefficient and write the corresponding evolution equation $L_t = [B_5, L]$.

(b) Prove that the resulting $B_5$ is formally skew-symmetric on compact smooth tests. Distinguish this from a statement about a closed skew-adjoint operator on $L^2$.

**Hint.** Collect coefficients of test derivatives in descending order, starting with the fourth derivative. Leibniz's rule makes this a finite triangular system. For the formal adjoint reverse each product before moving its derivatives.

**Answer.** The coefficients are $(-40, -60, -50, 30, -15, 30)$, giving

$$B_5 = 16D_x^5 - 40uD_x^3 - 60u'D_x^2 + (30u^2 - 50u'')D_x + 30uu' - 15u'''.$$

The scalar flow is $u_t = u^{(5)} - 10uu''' - 20u'u'' + 30u^2u'$. The operator is formally skew-symmetric on the stated test domain.

**Exercise 24.2 (Constancy for a sufficiently regular eigenbranch).** Let $u \in C^1(I; \mathcal{S}(\mathbb{R}, \mathbb{R}))$ solve $u_t = K_5[u]$ on an open interval $I$. Suppose $\lambda \in C^1(I; \mathbb{R})$ and $g \in C^1(I; H^7(\mathbb{R}, \mathbb{C}))$, with $g(t) \neq 0$ and $L(t)g(t) = \lambda(t)g(t)$ at every time. The Sobolev space $H^7$ uses the Fourier norm $\|\langle k\rangle^7 \widehat{g}\|_2$.

Use the following operator fact as a *given*: a real bounded multiplication perturbation of $-D_x^2$ is self-adjoint on the common domain $H^2(\mathbb{R})$. This is the relative-bound-zero case of Kato, *Perturbation Theory for Linear Operators*, V, Theorem 4.3, p. 287. Prove that $\lambda$ is constant. Specify the domain needed for moving $L$ across an inner product. Does this calculation require simplicity or isolation, and does it construct a branch or prove a theorem about the whole spectrum?

**Hint.** Differentiate the eigenvalue equation and pair with $g$. The coefficient identity from the first problem applies to the actual smooth eigenfunction once both ordered compositions belong to $L^2$.

**Answer.** The branch satisfies $\lambda' \|g\|_2^2 = (g, [B_5, L]g) = 0$. Here $B_5 g \in H^2$, so self-adjointness is applicable. No simplicity or isolation is required for this conditional computation. It does not supply eigenbranches or constancy of the entire spectrum.

**Exercise 24.3 (The same interior expression with moving endpoints).** Fix $\ell > 0$ and a smooth positive nonconstant function $\kappa(t)$. On $L^2(0,\ell)$ consider the expression $L = -\partial_x^2$ with domain

$$\mathcal{D}_t = \left\{ f \in H^2(0,\ell) : f(0) = 0, \quad f'(\ell) = \beta(t) f(\ell) \right\}, \qquad \beta(t) = \kappa(t)\coth(\kappa(t)\ell).$$

The trace values in this formula have their usual continuous Sobolev representatives on a finite interval.

(a) Show that this realization is symmetric by its endpoint Green formula. Find a nonzero smooth eigenfunction with a nonconstant eigenvalue. Self-adjointness is not required for this task.

(b) With $B = 0$, verify $L_t\psi = [B, L]\psi = 0$ on every compactly supported interior test. Explain why this does not contradict the preceding conditional eigenbranch result or establish constancy of the varying eigenvalue.

**Hint.** Try $\sinh(\kappa(t)x)$. The endpoint formula contains values of both functions and their first derivatives, not just the interior differential coefficients.

**Answer.** The eigenfunction is $\sinh(\kappa(t)x)$ with eigenvalue $-\kappa(t)^2$. The interior test identity is zero, but the domains $\mathcal{D}_t$ vary. The moving endpoint condition prevents using a single self-adjoint domain to cancel the differentiated-eigenfunction terms.